

CELLARHEAD, United Kingdom

WMO: 033380

Lat: 53.033N

Lon: 2.083W

Elev: 748

StdP: 14.30

Time Zone: 0.00 (EUW)

Period: 86-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.9	25.6	19.3	14.9	25.9	21.9	16.9	28.5	28.1	39.7	26.0	42.9	2.7	90	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	75.5	63.0	71.7	61.1	68.5	59.8	64.1	72.2	62.6	69.9	60.9	66.8	7.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	83.6	66.5	59.8	78.8	64.9	58.2	74.3	63.4	29.7	72.4	28.5	70.1	27.3	66.8	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	20.2	17.8	DB	17.2	80.3	5.0	4.7	13.5	83.6	10.6	86.4	7.8	89.0	4.1	92.4	(4)
(5)			WB	16.5	67.2	4.9	2.4	13.0	69.0	10.1	70.4	7.4	71.7	3.8	73.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.7	36.7	36.8	40.2	43.9	49.6	54.4	58.8	57.7	52.7	48.6	41.8	38.6	(6)	
	DBStd	9.38	5.61	6.07	4.92	5.27	5.60	5.37	4.61	5.14	4.40	5.10	5.77	5.92	(7)	
	HDD50	2090	411	371	306	197	76	15	0	2	21	87	250	354	(8)	
	HDD65	6697	876	791	770	634	477	320	200	235	370	510	696	818	(9)	
	CDD50	892	0	0	1	13	64	149	274	242	101	43	4	1	(10)	
	CDD65	21	0	0	0	0	0	3	8	10	0	0	0	0	(11)	
	CDH74	180	0	0	0	0	0	22	66	92	0	0	0	0	(12)	
	CDH80	26	0	0	0	0	0	2	4	20	0	0	0	0	(13)	
(14) Wind	WSAvg	7.9	9.1	9.7	10.3	8.2	7.8	6.8	6.9	6.5	6.4	7.9	7.6	7.6	(14)	
(15) Precipitation	PrecAvg	38.5	3.5	2.8	2.5	2.5	2.7	3.0	3.1	3.4	3.1	4.0	3.9	4.0	(15)	
	PrecMax	50.6	6.7	6.6	5.0	5.1	4.8	6.8	6.7	9.1	5.4	7.9	7.9	7.4	(16)	
	PrecMin	29.0	0.6	0.6	0.4	0.4	0.6	1.0	1.1	0.9	0.6	1.3	1.6	1.1	(17)	
	PrecStd	5.7	1.6	1.3	1.3	1.3	1.1	1.6	1.3	1.5	1.3	1.7	1.5	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	51.5	55.8	65.0	71.7	76.9	79.4	82.6	70.6	63.6	56.2	54.0	(19)	
		MCWB	49.1	48.2	50.0	55.5	59.7	63.8	63.0	63.3	61.5	57.9	53.8	51.7	(20)	
	2%	DB	48.8	49.4	51.9	59.8	67.3	72.0	75.5	75.2	65.6	59.9	54.3	52.3	(21)	
		MCWB	46.7	46.7	48.2	52.7	57.7	60.8	62.9	63.0	58.6	55.6	52.5	50.8	(22)	
	5%	DB	46.9	47.7	49.9	55.5	63.0	68.4	71.7	70.2	62.3	57.8	52.2	49.7	(23)	
		MCWB	45.0	45.4	47.1	50.0	54.9	59.6	61.4	61.0	56.4	54.3	50.4	48.4	(24)	
	10%	DB	45.1	45.8	47.8	52.5	59.1	64.8	68.4	66.4	60.2	56.0	50.5	47.9	(25)	
		MCWB	43.4	43.7	45.2	47.9	52.6	57.5	59.9	59.3	55.4	53.3	49.0	46.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	49.3	51.1	56.4	61.4	65.9	66.2	65.6	62.3	59.0	54.9	52.5	(27)	
		MCDB	50.6	50.7	54.3	62.3	69.6	70.5	73.7	75.2	69.3	62.2	55.4	53.6	(28)	
	2%	WB	47.3	47.4	49.2	53.1	58.4	62.8	64.3	63.8	59.9	56.8	52.6	50.8	(29)	
		MCDB	48.5	48.8	51.2	59.0	65.7	70.5	72.9	72.3	63.7	58.9	53.8	52.1	(30)	
	5%	WB	45.4	45.8	47.3	50.9	55.8	60.2	62.6	62.0	58.1	55.2	50.7	48.5	(31)	
		MCDB	46.6	47.4	49.2	54.7	61.7	66.8	69.3	69.0	61.2	56.9	52.0	49.5	(32)	
	10%	WB	43.6	43.9	45.7	48.7	53.5	58.0	60.9	60.3	56.3	53.8	49.2	46.5	(33)	
		MCDB	44.9	45.6	47.4	51.9	58.5	64.1	66.9	65.6	59.4	55.5	50.4	47.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.8	10.0	12.1	13.9	13.7	13.7	14.1	12.4	10.5	8.8	7.4	(35)	
		MCDBR	10.6	10.2	11.9	15.5	19.5	20.7	21.1	20.3	15.2	12.4	9.1	8.6	(36)	
	5% WB	MCWBR	10.1	9.0	9.0	9.4	11.6	11.8	10.8	9.9	9.5	9.1	8.3	8.6	(37)	
		MCWBR	10.3	10.3	10.6	15.0	17.1	17.7	17.7	17.6	13.5	10.3	9.0	8.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.315	0.330	0.360	0.385	0.388	0.387	0.387	0.373	0.358	0.340	0.316	0.304	(40)	
		taud	2.350	2.397	2.316	2.256	2.285	2.312	2.314	2.373	2.419	2.445	2.408	2.324	(41)	
	Ebn at Noon		210	246	260	268	273	274	272	269	259	239	209	189	(42)	
		Edh at Noon	19	24	32	38	39	39	38	34	29	23	18	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		218	432	784	1210	1517	1597	1521	1249	921	521	265	170	(44)	
	RadStd		20	49	84	129	100	156	159	86	76	45	29	18	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page